
Sheet 6

Question 1

A DWDM optical transmission system is designed to have 100-GHz channel spacings. How many wavelength channels can be utilized in the 1536-to-1556-nm spectral band?

Question 2

A four channel CWDM system using laser diodes has the optical parameters shown in the following table. The system operates over 10 km. What is the system operating margin? Assume connector losses at the multiplexer source, and detector are 1.0 dB.

Source Wavelength (nm)	1551	1571	1591	1611
Source output (dBm)	0.0	-1.0	-0.05	-1.8
Multiplexer insertion loss (dB)	2.9	3.5	4.1	3.1
Fiber attenuation (dB/km)	0.25	0.37	0.42	0.47
Demultiplexer insertion loss (dB)	3.6	3.1	3.6	3.8
Receiver Sensitivity (dBm)	-56	-55.5	-55.9	-55

Question 3

It is required to design an optical communication link at the speed of 2 GB/Sec. NRZ digital signal between Cairo and Aswan separated by a distance of about 850 km and operating at the two wavelengths 1310 nm and 1550 nm simultaneously. The optical components available have the following spec's at the operating wavelength:

- Single mode fiber (SMF) with attenuation $\alpha = 0.25$ dB/km at 1550 nm and 0.45 dB/Km at 1310 nm
- Dispersion parameter $D = 2$ ps/(km.nm) at 1550 nm and zero at 1310 nm
- Optical Receiver with Sensitivity = -50 dBm at 1550 nm and -55 dBm at 1310 nm and a Band Width BW = 5 GHz
- Two laser transmitters are available, the first is a laser source at 1310 nm with output power of 10 mW in the SMF fiber and a spectral width of $\Delta\lambda = 1$ nm. This laser transmitter has a rise time of 0.2 nsec. The second is a Laser transmitter at the wavelength of 1550 nm with an output power in the SMF = 1 mW, a rise time of 0.1 nsec., and a spectral width of $\Delta\nu = 10$ GHz.
- Mux/Demux unit Insertion loss=3 dB
- All the components are terminated with connectors and the average connector loss is 0.5 dB per connection (fiber to fiber loss).

Calculate the maximum distance "L" that can be achieved without repeater (the repeater_less distance) and thus determine the number of repeaters required for implementing the system.

(assume system margin of 6 dB)

Question 4

It is required to design a low cost FTTH optical distribution network using the configuration shown in the following figure below to serve 15 customers. As an FTTH network, it will be working at the wavelengths of 1490 nm for the data download, 1310 nm for the data upload and 1550 nm for the video distribution (download). At these wavelengths, the available components have the following properties:

Wavelength	Laser source			Detector		SNR required	Fiber Losses
	Power	Rise time	Spectral width	Sensitivity	BW		
1310 nm	4 mW	0.05 nsec	0.1 nm	-50 dBm	10 GHz	0dB	0.45 dB/Km
1490 nm	10 mW	0.1 nSec	0.1 nm	-46 dBm	10 GHz	0 dB	0.5 db/Km
1550 nm	4 mW	0.05 nsec	0.05 nm	-50 dBm	10 GHz	6 dB	0.25 dB/ km

- Assuming a system power margin of 6 dB, calculate the maximum length that can be covered by this network
- If the SM fiber used in the network has a chromatic dispersion parameter D that can be expressed by:

$$D(\lambda) = \frac{S_0}{4} \left[\lambda - \frac{\lambda_0^4}{\lambda^3} \right]$$

where $\lambda_0 = 1317$ nm is the Zero Dispersion wavelength and $S_0 = 0.088$ ps/(nm²·km) is the dispersion slope, what is the max. bit rate that can be transmitted on the system at each wavelength. Assume a RZ format for the line code and that each splitter has practically 3.5 dB loss

